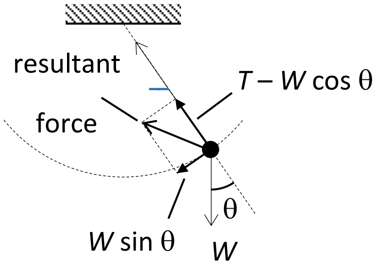
Power lost due to air resistance = air resistance x velocity = 200 x 40 = 8 MW

Rate at which kinetic energy is increasing = 20 – 8 = 12 MW

9 B



As the bob undergoes circular motion, there must be a resultant force acting on the bob in radial direction = centripetal force

$= T - W \cos \theta$

In the tangential direction, resultant force acting on the bob = $W \sin \theta$

Hence, the resultant force should be in direction Q.

10 A At the bottom of the hill, the net force = N – mg = centripetal force = ma